

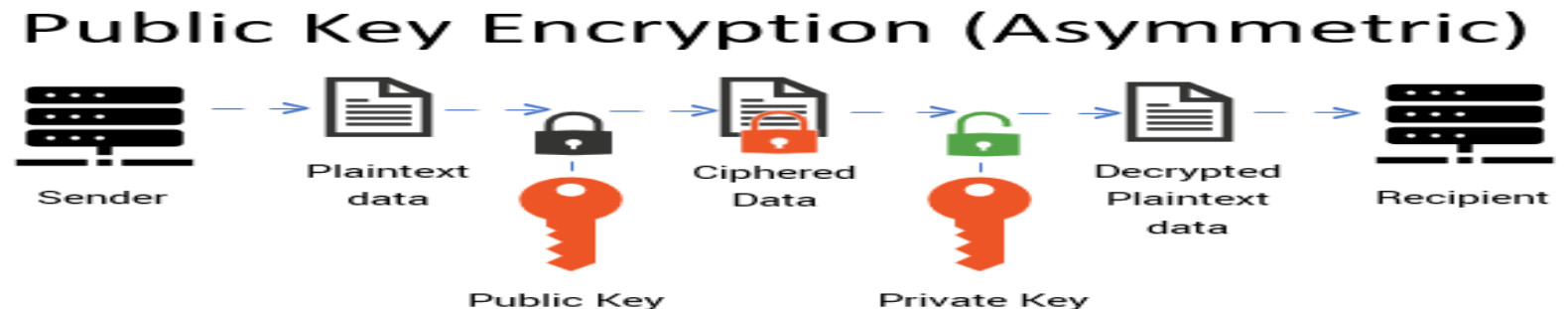
Cyber Security & Laws

BCAC 0032

Module-II

Public key cryptography

- In 1976, Whitfield Diffie and Martin Hellman introduced the concept of Public Key cryptography (asymmetric cryptography).
- In their system, each person is the owner of a mathematically related pair of keys: a Public Key, intended to be available to anyone who wants it; and a Private Key, which is kept secret and only known by the owner.
- Because messages are encrypted with a Public Key and can only be decrypted by the related Private Key.
- In contrast to symmetric key systems, public key systems require each user **A** to have two keys: a public key, $K_{pub}(A)$ that is published and a private key $K_{pri}(A)$ that is kept secret. A message encrypted (with a ruleset **E**) using one key can be decrypted (using ruleset **D**) using only the other key.



Public key Encryption Algorithms

- **RSA Cryptosystem**
- **Diffie-Hellman Key Exchange**
- **Elliptic Curve Cryptography**

RSA Algorithms

- RSA is the best known, and by far the most widely used general public key encryption algorithm, and was first published by Rivest, Shamir & Adleman of MIT in 1978.
- RSA involves a public-key and a private-key where the public key is known to all and is used to encrypt data or message. The data or message which has been encrypted using a public key can only be decrypted by using its corresponding private-key. Each user generates a key pair i.e. public and private key using the following steps:
 - Selects two large primes at random - p, q
 - compute the modulus $n=p.q$
 - calculate $\phi(n)$, where $\phi(n)=(p-1)(q-1)$
 - selecting at random the encryption key e , where $1 < e < \phi(n)$, and $\gcd(e, \phi(n))=1$
 - solve following equation to find decryption key d : $e.d=1 \pmod{\phi(n)}$ and $0 \leq d \leq n$
 - publish their public encryption key: $KU=\{e,n\}$
 - keep secret private decryption key: $KR=\{d,n\}$

- Both the sender and receiver must know the values of n and e , and only the receiver knows the value of d . Encryption and Decryption are done using the following equations.
- To encrypt a message M the sender:
 - obtains public key of recipient $KU=\{e,n\}$
 - computes: $C=M^e \bmod n$, where $0 \leq M$
- To decrypt the ciphertext C the owner:
 - uses their private key $KR=\{d,n\}$
 - computes: $M=C^d \bmod n = (M^e)^d \bmod n = M^{ed} \bmod n$

Example Calculation

1. **Choose Primes:** $p = 7, q = 19$.
2. **Calculate n :** $n = pq = 7 \times 19 = 133$.
3. **Calculate $\phi(n)$:** $\phi(n) = (p - 1)(q - 1) = 6 \times 18 = 108$.
4. **Choose e :** Choose e such that $1 < e < 108$ and $\gcd(e, 108) = 1$. Let $e = 5$.
5. **Calculate d :** d such that $5d \equiv 1 \pmod{108}$. A possible value for d is 65 (since $5 \times 65 = 325$, and $325 \pmod{108} = 1$).
6. **Encryption:** $C = M^e \pmod{n}$.
7. **Decryption:** $M = C^d \pmod{n}$. 

Deffie-Hellman Key Exchange

- The Diffie-Hellman (DH) algorithm is a foundational cryptographic protocol (1976) that allows two parties to securely establish a shared secret key over an insecure communication channel, without transmitting the key itself.
- It enables secure, symmetric encrypted communication between parties that have no prior knowledge of each other.

Diffie-Hellman Key Exchange Algo

1. **Public Parameters:** Alice and Bob publicly agree on a large prime number p and a primitive root modulo p , g .
2. **Private Keys:** Alice chooses a secret private key a . Bob chooses a secret private key b .
3. **Public Key Calculation:**
 - Alice computes public key $A = g^a \pmod{p}$.
 - Bob computes public key $B = g^b \pmod{p}$.
4. **Exchange:** Alice sends A to Bob, and Bob sends B to Alice.
5. **Shared Secret Generation:**
 - Alice computes shared secret $K = B^a \pmod{p} = (g^b)^a \pmod{p}$.
 - Bob computes shared secret $K = A^b \pmod{p} = (g^a)^b \pmod{p}$.

Both parties now possess the same secret $K = g^{ab} \pmod{p}$

Numerical

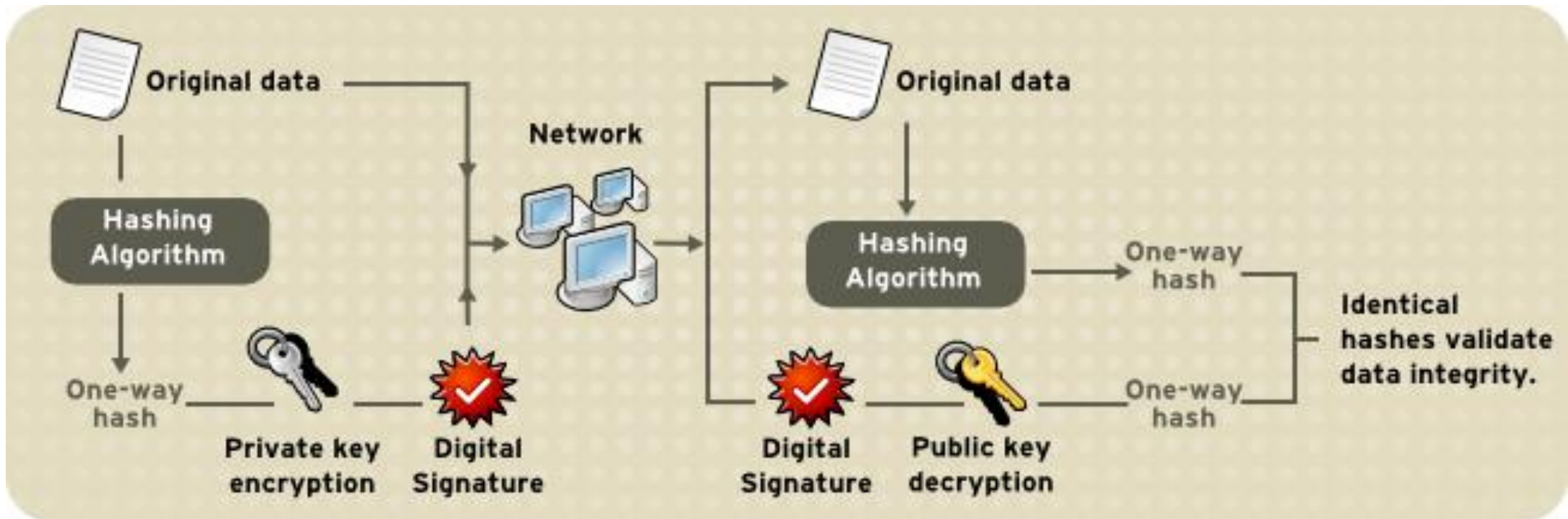
- **Public Values:** $p = 23, g = 5$
- **Private Values:** Alice $a = 4$, Bob $b = 3$
- **Public Exchange:**
 - Alice sends $A = 5^4 \pmod{23} = 625 \pmod{23} = 4$
 - Bob sends $B = 5^3 \pmod{23} = 125 \pmod{23} = 10$
- **Secret Calculation:**
 - Alice computes $K = B^a \pmod{p} = 10^4 \pmod{23} = 18$
 - Bob computes $K = A^b \pmod{p} = 4^3 \pmod{23} = 18$
- **Result:** Shared Secret = 18. 

Applications of Public-key Cryptosystems

- **Authentication-** Authentication is the process of verifying one's identity.
- **Key Exchange-** The requirement is to share a key, which can later be used to encrypt and decrypt messages by a secret key system like DES.
- **Confidentiality-** Confidentiality means protecting data from eavesdroppers.
- **Non-repudiation-** This requirement comes from the issue of denying previous actions.
- **Digital Signatures-** The concept is much similar to the real world process of signing a document to make it legal and to specify the person responsible for the document.
- **Data Integrity-** Digital signatures can also be used to validate the integrity of signed data.

Digital Signature

A digital signature is a cryptographic mechanism using [asymmetric cryptography](#) to verify the authenticity, integrity, and non-repudiation of digital messages or documents.



Working of Digital Signature

- **How it Works:** The sender hashes the document and encrypts the hash with their private key, creating a unique signature. The recipient decrypts this with the sender's public key and verifies the hash matches.
- **Key Components:**
 - **Hash Function:** Produces a unique digital fingerprint of the data.
 - **Private key:** Used by the sender to sign the hash.
 - **Public key:** Used by anyone to verify the signature.

Applications of DS

1. Secure Email Communication

Digital signatures are used in email systems to verify the sender's identity and ensure that the message has not been modified during transmission. Example: Secure email protocols like **S/MIME** and **PGP** use digital signatures.

2. E-Commerce Transactions

In online business transactions, digital signatures help authenticate buyers and sellers and secure payment information. They ensure **secure contracts, invoices, and purchase orders** in online marketplaces.

3. Electronic Documents and Contracts

Digital signatures are commonly used to sign **PDF documents, agreements, and legal contracts** electronically. They eliminate the need for physical signatures and enable faster document processing.

4. E-Governance Services

Government services use digital signatures for secure digital transactions. Examples include: Filing **income tax returns**

E-tendering and e-procurement

Online submission of official forms

5. Software Distribution

Software companies digitally sign applications and updates to prove that the software comes from a trusted source and has not been tampered with.

6. Banking and Financial Transactions

Digital signatures help secure online banking operations such as: Fund transfers, Digital payment authentication, Secure financial documents

7. Blockchain and Cryptocurrencies

Digital signatures are used to verify transactions in blockchain networks. They ensure that only the **owner of a private key** can authorize a transaction.

8. Secure Data Transmission

Digital signatures protect sensitive data during transmission in **military, healthcare, and corporate networks** by verifying the source and integrity of information.

9. Electronic Voting Systems

Digital signatures help authenticate voters and maintain the integrity of electronic voting processes.

10. Healthcare Systems

In healthcare, digital signatures are used to sign: Electronic health records (EHR), Medical prescriptions, Clinical reports

Digital watermarking

- **Digital watermarking** is a technique used to embed hidden information (called a **watermark**) into digital media such as :
 - images,
 - audio,
 - video, or
 - documents.
- The embedded information is used to **verify ownership, authenticate content, and prevent unauthorized copying or distribution.**
- The watermark is usually invisible to users but can be detected or extracted using specific algorithms.

Features/characteristics of Digital Watermarking

- **1. Imperceptibility** – The watermark should not affect the quality of the original content.
- **2. Robustness** – The watermark should remain detectable even after compression, editing, or noise.
- **3. Security** – Unauthorized users should not be able to remove or modify the watermark.
- **4. Capacity** – Ability to embed sufficient information without degrading the content.

- **5. Reliability**

- The watermark detection system should be **accurate and reliable**, meaning it should correctly identify the presence or absence of the watermark.

- **6. Computational Efficiency**

- The watermark embedding and extraction processes should require **reasonable computational resources** so that they can be applied in real-time applications.

- **7. Unambiguity**

- The watermark should **clearly identify the owner or source** of the digital content without confusion or false detection.

Types of Digital Watermarking

- **1. Visible Watermarking**

- The watermark is **visible to users**.
- Commonly used in images and videos to indicate ownership.
- Examples:
 - TV channel logos on broadcasts
 - Watermarks on stock photos

- **2. Invisible Watermarking**

- The watermark is **hidden within the media** and cannot be seen by the human eye.
- It is detected using special software.
- Used for copyright protection and content authentication.

3. Fragile Watermarking

Designed to detect **tampering or modification** in digital content.

If the content is changed, the watermark becomes invalid.

Used in **authentication systems**.

4. Robust Watermarking

Designed to survive **compression, resizing, filtering, and noise**.

Mainly used for **copyright protection**.